

Total number of printed pages – 4

23R-D25 (PHY5404C)

2025

PHYSICS

(Major – IV)

Paper : PHY5404C

(Research Methodology)

Full Marks : 60

Time : 2½ hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions : $1 \times 7 = 7$
- ✓(a) State one tool used for statistical analysis in scientific writing.
 - (b) Mention *any two* key features of scientific writing.
 - ✓(c) Name two citation management tools used in scientific writing.
 - ✓(d) Name *any one* software used for collaborative LaTeX editing.
 - (e) What does FFP stand for in the context of scientific misconduct ?

(f) What is the difference between primary and secondary sources of literature review?

(g) Define Intellectual Property Rights (IPR).

2. Answer the following questions : $2 \times 4 = 8$

(a) Explain the role of the literature review in a scientific paper.

(b) What is meant by "citation integrity" in scientific writing?

(c) Differentiate between *descriptive* and *analytical* research.

(d) State *two* qualities of *intellectual honesty* in research.

3. (a) Draw and explain the hierarchical diagram for data classification. 5

(b) Evaluate the role of referencing in validating new knowledge. 5

Or

(c) Discuss the ethical and compliance considerations in preparing a scientific document.

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- (d) Write briefly on the main criteria of good research. 5

Or

- (e) Explain the significance of *research design* in conducting research.

4. (a) What are the different steps involved in treating data in a scientific study? Explain each step in simple terms. 10

Or

- (b) Explain the modular structure of a standard document. Briefly describe the function of each section and justify why this structure is essential for effective scientific communication.

5. (a) Discuss the characteristics and different types of research with suitable examples. 10

Or

- (b) Explain the *process of problem identification, formulation, and evaluation* of research.

6. (a) What are Intellectual Property Rights (IPR)? Explain their importance and discuss the role of *patent laws*, *copyright*, and *cyber laws* in protecting research findings. 10

Or

- (b) Compare fundamental research with action research by giving suitable examples. Write short notes on plagiarism and salami slicing with examples.